

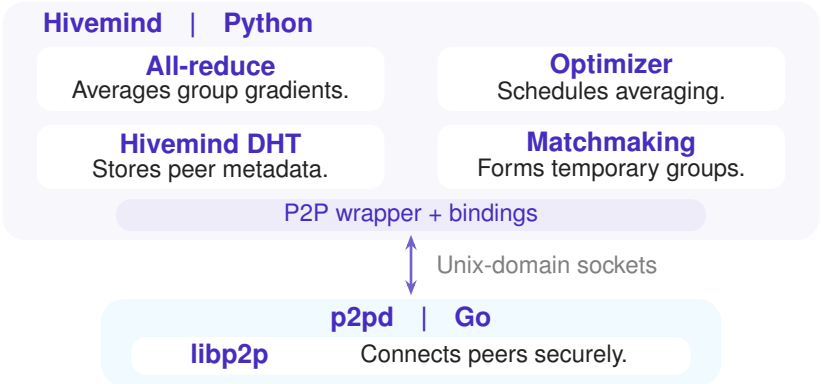
Hivemind

Sarat Yellapragada

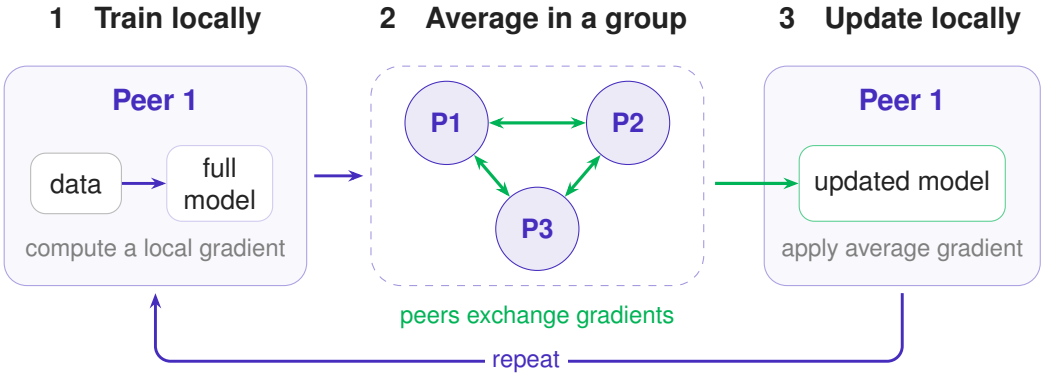
Secura Lab

Chalmers University of Technology | 18 September 2026

Inside a Hivemind peer

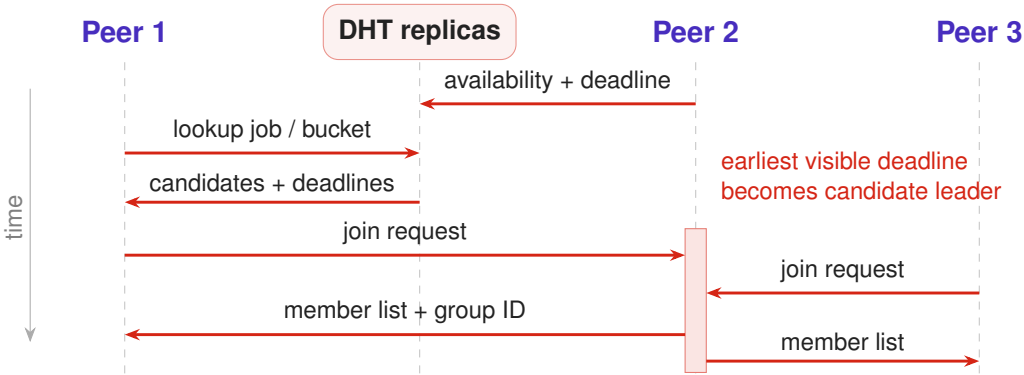


Hivemind's training loop



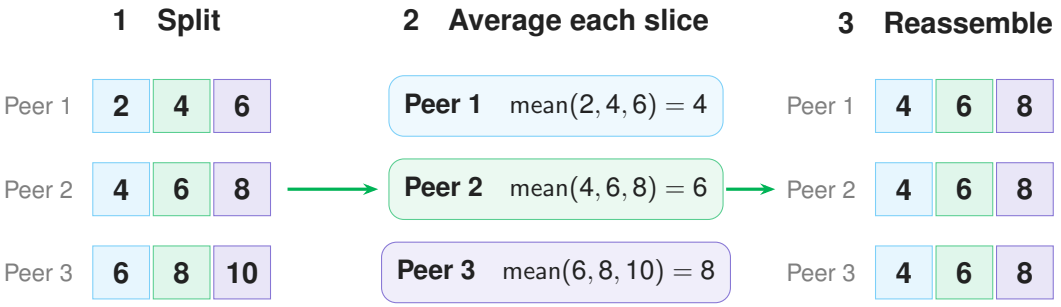
Every peer keeps its own data and full model | no central aggregation server

How a group forms



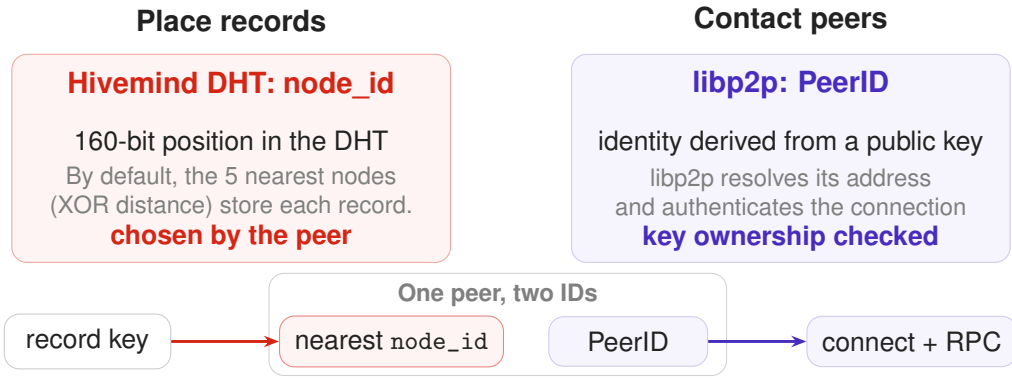
The temporary leader checks requests and admits members.

All-reduce: averaging within the group



All-reduce: every group member obtains the aggregate | equal-weight illustration

Two IDs



The routing table stores both IDs; libp2p connects using PeerID.

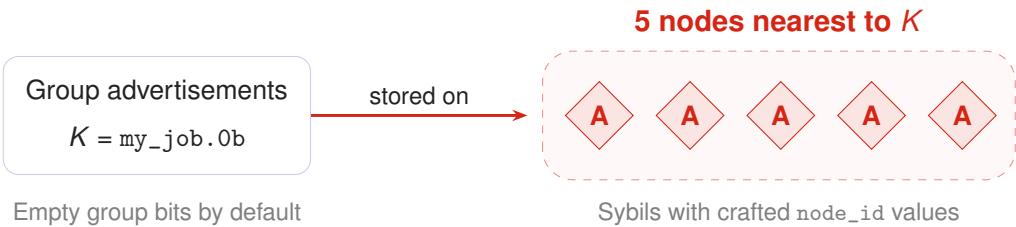
Questions?

1. Become the record holders

Known group discovery key
`<prefix>.0b<group_bits>`

+

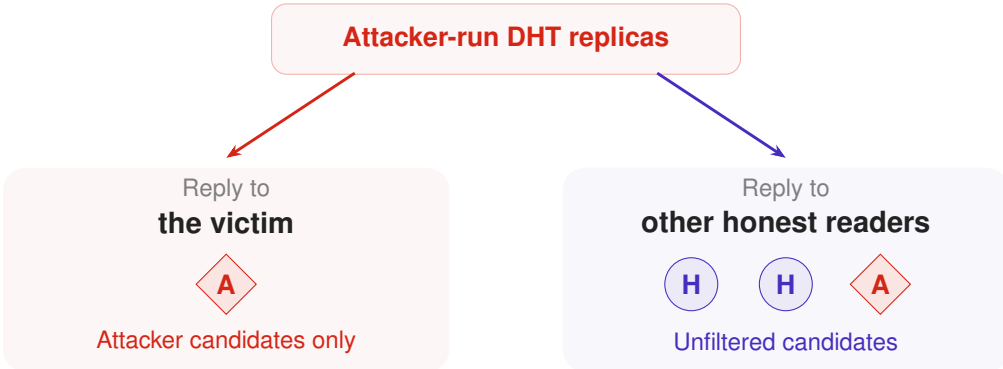
Self-chosen `node_id`



All five replicas sit on attacker-run nodes.

▸ Exact tables

2. Give the victim a different view



Goal: steer the victim into an attacker-led group.

▸ Exact tables

β is chosen, not inherited

When the attacker leads, it admits only the victim + its own peers.

1 attacker

$\alpha = 1/16$ of the swarm



$\beta = 0.500$

2 peers in the group

2 attackers

$\alpha = 2/16$ of the swarm

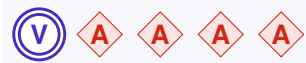


$\beta = 0.667$

3 peers in the group

4 attackers

$\alpha = 4/16$ of the swarm



$\beta = 0.800$

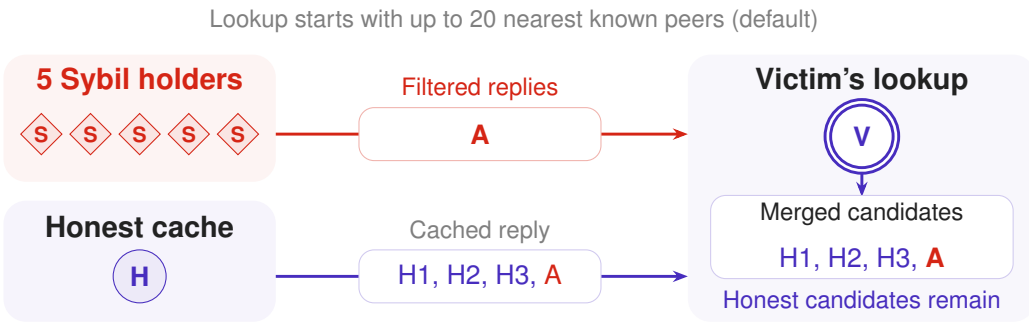
5 peers in the group

Target group size: 8 | minimum: 2 (default)

Even 50% attacker share is outside Krum's guarantee.

▸ Exact tables

Caching



5 Sybils: 0% capture **20 Sybils: 100% capture**

Capture rates: separate empty-reply experiment | caching on

▸ Exact tables

Capture persisted across rounds

8 training peers (3 attackers) + 25 DHT-only Sybils

Run	Captured rounds	Recorded rounds
1	**15**	15
2	**13**	13
3	**12**	12

All 40 recorded rounds: victim + attacker peers only.

Sybils joined before training | caching on | sharding off

16 training peers 3 attempts | 0 victim rounds Capture not assessable

Arrival-to-capture delay unmeasured | 16-peer attempts: handoff report

▸ Exact tables

Tracking the bucket, chasing the group

Victim rounds with at least one attacker in the group

Training peers	Fixed bucket	Tracks victim
16	56%	**84%**
32	40%	**72%**
48	18%	**38%**

A correct bucket prediction does not guarantee the same averaging group.

Next-bucket predictions: 63/63 correct, using shared group information.

Sharding on | 4 discovery buckets | 4 attackers | 2 runs per setting

▸ Exact tables

Poisoning: results and limits

8 training peers | 1 poisoner | 64-parameter regression

- One peer reversed and scaled its outgoing gradients, then claimed an averaging weight of one million.
- Weighted mean left honest peers farther from the optimum than in matching clean runs.
- Coordinate median substantially reduced the damage across all three seeds.

Poisoning was tested separately from group capture.
Krum: theoretical limits only; no implementation or experiment.

▸ Exact tables

Limitations

	Tested setting	Limit of the evidence
Model	Linear regression 64 parameters	No transformer experiments
Network	8–104 nodes on one machine	Large-network routing untested
Timing	Sybils joined before training	Time to establish capture not measured
Attack	Stages tested separately	Full attack chain unmeasured
Krum	Theoretical limits only Not implemented	Needs full-gradient distances; reducers receive slices

Control-plane defences are untested; default group mixing remains unsettled.

▸ All experiment tables